

**<sup>67</sup>Cu - Comments on evaluation of decay data**

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<sup>67</sup>Cu is a promising radionuclide for cancer diagnostics and radio-immunotherapy [2002Nova]. The  $\beta^-$  branching intensities in <sup>67</sup>Cu decay are also of interest in the measurement of the  $\beta$ -asymmetry parameter, which can provide information on the search for new physics effects beyond the Standard Model [2014So16].

**1 Nuclear data**

The <sup>67</sup>Cu radionuclide decays by  $\beta^-$  emission to several <sup>67</sup>Zn levels. The Q-value has been adopted from AME2020 [2021Wa16]. The level energies have been established from a least-square fitting of the evaluated  $\gamma$ -ray energies, as detailed in section 1.3.1. The spins and parities of the <sup>67</sup>Zn levels and of the <sup>67</sup>Cu ground state are well established. They have been adopted from [1966Fr12].

Standardization with the  $4\pi\beta$ (Proportional Counter)- $\gamma$  coincidence method was reported in [2018Saha]. Further improvement of the decay scheme would require new precise measurements of the <sup>67</sup>Cu half-life and of the absolute  $\gamma$ -ray emission intensities.

**1.1. Half-lives****1.1.1. Half-life of the <sup>67</sup>Cu ground state**

Table 1 summarizes the measured half-lives of <sup>67</sup>Cu decay available in the literature. Uncertainties not reported at  $1\sigma$  have been corrected.

Table 1 – Measured half-lives of <sup>67</sup>Cu decay.

| Reference      | T <sub>1/2</sub> (h) | Comment                                                                              |
|----------------|----------------------|--------------------------------------------------------------------------------------|
| 1956Ru45       | 61.6 (3)             |                                                                                      |
| 1961Wi19       | 61.54 (15)           |                                                                                      |
| 1968Re04       | 61.88 (14)           |                                                                                      |
| 1970Ma56       | 61.00 (25)           | Original uncertainty of 0.5 given at $2\sigma$ , excluded by the Chauvenet criterion |
| 1972La14       | 61.83 (7)            | Original uncertainty of 0.21 given at $3\sigma$                                      |
| 1974Ro18       | 62.01 (14)           |                                                                                      |
| 2011Ko49       | 61.81 (22)           |                                                                                      |
| 2014So16       | 61.07 (12)           | Lower half-life due to pileup effect, excluded                                       |
| <b>Adopted</b> | <b>61.82 (5)</b>     | <b>Weighted mean with internal uncertainty</b>                                       |

The full dataset is discrepant ( $\chi^2 = 6.67$  and  $\chi^2_{\text{crit.}} = 2.64$ ) due to the most recent value from [2014So16]. The authors mention that their measured half-life is lower than other values in the literature due to uncorrected pileup. This value has therefore been excluded. The value from [1970Ma56] is then excluded by the Chauvenet criterion. The resulting dataset is consistent ( $\chi^2 = 1.21$  and  $\chi^2_{\text{crit.}} = 3.02$ ). The adopted value of  $T_{1/2} = 61.82$  (5) h or 2.5758 (21) d corresponds to the weighted average with internal uncertainty. The value from [1972La14] takes 53% of the weight.

### 1.1.2. Half-life of the $^{67}\text{Zn}$ 93-keV level

Table 2 summarizes the measured half-lives of the  $^{67}\text{Zn}$  93-keV level available in the literature. Some values have been excluded for various reasons: in the absence of reported uncertainty [1953Ke51]; superseded by a more recent value [1972Le37]; reported with a too large uncertainty [1953Me52, 1969lv02]; and excluded by the Chauvenet criterion. The value from [1973Le18] results from the average of two measurements, one from  $^{67}\text{Cu}$  decay and the other from  $^{67}\text{Ga}$  decay. The uncertainties reported at  $3\sigma$  have been corrected.

The resulting dataset is discrepant ( $\chi^2 = 3.30$  and  $\chi^2_{\text{crit.}} = 3.02$ ) due to the value from [1996Hw03]. The dataset becomes consistent ( $\chi^2 = 2.53$  and  $\chi^2_{\text{crit.}} = 3.02$ ) when the uncertainty of [1996Hw03] is increased from 0.03 to 0.04, which can be justified by the fact that the reported uncertainty most likely comes only from the least-square fitting. The adopted value of  $T_{1/2}(93\text{-keV}) = 9.130$  (18)  $\mu\text{s}$  corresponds to the weighted average with internal uncertainty. The value from [2014Di03] takes 72% of the weight.

Table 2 – Measured half-lives of  $^{67}\text{Zn}$  93-keV level.

| Reference      | $T_{1/2}$ ( $\mu\text{s}$ ) | Comment                                                                                                                                     |
|----------------|-----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| 1953Ke51       | 8.5                         | Excluded, no uncertainty                                                                                                                    |
| 1953Me52       | 9.5 (10)                    | Excluded, uncertainty too large                                                                                                             |
| 1957Bu39       | 9.4 (3)                     |                                                                                                                                             |
| 1969lv02       | 8.8 (18)                    | Excluded, uncertainty too large                                                                                                             |
| 1971Su18       | 8.7 (1)                     | Excluded by the Chauvenet criterion                                                                                                         |
| 1972Le37       | 9.1 (15)                    | Excluded, same authors as [1973Le18]                                                                                                        |
| 1973Le18       | 9.15 (7)                    | From $^{67}\text{Cu}$ , 9.10(20) $\mu\text{s}$ and $^{67}\text{Ga}$ , 9.20(20) $\mu\text{s}$ .<br>Original uncertainties given at $3\sigma$ |
| 1975Ro25       | 9.1 (4)                     |                                                                                                                                             |
| 1996Hw03       | 9.01 (4)                    | Uncertainty increased from 0.03 to 0.04 to<br>obtain a consistent dataset                                                                   |
| 1998At04       | 9.34 (20)                   |                                                                                                                                             |
| 2014Di03       | 9.157 (21)                  |                                                                                                                                             |
| <b>Adopted</b> | <b>9.130 (18)</b>           | <b>Weighted mean with internal uncertainty</b>                                                                                              |

### 1.1.3. Half-life of the $^{67}\text{Zn}$ 184-keV level

Table 3 summarizes the measured half-lives of the  $^{67}\text{Zn}$  184-keV level available in the literature. The values from [1961Ho05, 1962Ri09] have been excluded by the Chauvenet criterion. The original

uncertainty reported in [1964Al28] as  $1.10^{(+0.51}_{-0.48)}$  ns has been symmetrized. The resulting dataset is consistent ( $\chi^2 = 3.10$  and  $\chi^2_{\text{crit.}} = 3.32$ ). The adopted value of  $T_{1/2}(184\text{-keV}) = 1.059$  (16) ns corresponds to the weighted average with external uncertainty. The weight of the value from [2022Ms01] has been limited to 50%.

Table 3 – Measured half-lives of  $^{67}\text{Zn}$  184-keV level.

| Reference      | $T_{1/2}$ (ns)    | Comment                                        |
|----------------|-------------------|------------------------------------------------|
| 1961Ho05       | 1.45 (15)         | Excluded by the Chauvenet criterion            |
| 1962Ri09       | 2.2 (7)           | Excluded by the Chauvenet criterion            |
| 1964Al28       | 1.1 (5)           | Original uncertainty symmetrized               |
| 1968Li02       | 1.01 (5)          |                                                |
| 1972En08       | 1.026 (14)        |                                                |
| 1975Ro25       | 1.06 (4)          |                                                |
| 2022Ms01       | 1.090 (7)         | Weight limited to 50%                          |
| <b>Adopted</b> | <b>1.059 (16)</b> | <b>Weighted mean with external uncertainty</b> |

## 1.2. Beta transitions

The four  $\beta^-$  transitions of the decay scheme are of allowed nature. Table 4 summarizes their evaluated properties. The transition probabilities have been deduced from the balance of the  $\gamma$  transition probabilities determined in section 1.3.5. The sum of the probabilities has been constrained to 100%. However, once the probabilities are rounded considering their uncertainties, the apparent sum is only 99.969%. The BetaShape program [2023Mo21] has been applied on the  $\beta^-$  transitions to determine the average energies of the continuous spectra and the  $\log ft$  values.

Table 4 – Properties of the  $\beta^-$  transitions in  $^{67}\text{Cu}$  decay.

| Transition    | Energy (keV) | Probability (%) | Rounded probability (%) | Nature  | $\log ft$  |
|---------------|--------------|-----------------|-------------------------|---------|------------|
| $\beta_{0,3}$ | 167.3 (8)    | 1.039 (14)      | 1.039 (14)              | Allowed | 5.771 (9)  |
| $\beta_{0,2}$ | 376.2 (8)    | 51.735 (628)    | 51.7 (6)                | Allowed | 5.224 (6)  |
| $\beta_{0,1}$ | 467.5 (8)    | 19.734 (470)    | 19.73 (47)              | Allowed | 5.965 (11) |
| $\beta_{0,0}$ | 560.8 (8)    | 27.492 (784)    | 27.5 (8)                | Allowed | 6.098 (13) |

## 1.3. Gamma transitions

### 1.3.1. Gamma-ray energies

Table 5 summarizes the measured  $\gamma$ -ray energies emitted in  $^{67}\text{Cu}$  decay available in the literature. The values from [1978Me10] are superseded by those from [1990Me15]. The values in [1974HeYW] were measured from both  $^{67}\text{Cu}$  and  $^{67}\text{Ga}$  decays. A weighted mean of the two values with minimum experimental uncertainty have been considered for each  $\gamma$ -ray energy. Some values marked in Table 5 have been excluded by the Chauvenet criterion.

For each  $\gamma$ -ray energy, the dataset is consistent ( $\chi^2$  and  $\chi^2_{\text{crit.}}$  are provided in Table 5). The evaluated energies correspond to the weighted averages with internal uncertainties, except for  $\gamma_{2,1}$

for which external uncertainty has been adopted. Table 6 summarizes the evaluated  $\gamma$ -ray energies and the corresponding recoil energies, calculated using the atomic mass from [2021Wa16].

Table 5 – Measured  $\gamma$ -ray energies emitted in  $^{67}\text{Cu}$  decay.

| Reference                      | $\gamma_{2,1}$     | $\gamma_{1,0}$     | $\gamma_{2,0}$      | $\gamma_{3,2}$      | $\gamma_{3,1}$      | $\gamma_{3,0}$      |
|--------------------------------|--------------------|--------------------|---------------------|---------------------|---------------------|---------------------|
| 1953Ea11                       |                    | 92                 | 182                 |                     |                     |                     |
| 1953Ke51                       | 90                 | 92                 | 182                 | 206                 | 296                 | 388                 |
| 1953Nuss                       |                    | 94                 | 190                 |                     | 304                 | 393                 |
| 1954Bo97                       |                    | 98                 | 186                 |                     |                     |                     |
| 1957Be59                       |                    |                    | 182                 | 296                 |                     | 388                 |
| 1958Ch08                       | 91.22 (4)          | 93.26 (4)          | 184.46 (27)         |                     |                     |                     |
| 1966Fr12                       | 91.275 (20)        | 93.317 (20)        | 184.595 (40)        | 208.959 (60)        | 300.242 (65)        | 393.65 (6)          |
| 1969Ra15                       | 91.26 (10)         | 93.25 (10)         | 184.53 (10)         | 208.95 (10)         | 300.22 (10)         | 393.6 (1)           |
| 1971Su18                       | 91 (2)*            | 93 (2)*            | 184 (2)             | 208 (2)*            | 299 (2)*            | 393 (2)             |
| 1974Ar22                       |                    | 93.2 (2)*          | 184.0 (2)           |                     |                     |                     |
| From $^{67}\text{Cu}$          | 91.31 (3)          | 93.32 (2)          | 184.56 (2)          | 208.93 (2)          | 300.24 (6)          | 393.56 (7)          |
| From $^{67}\text{Ga}$          | 91.31 (5)          | 93.32 (2)          | 184.56 (2)          | 208.93 (2)          | 300.18 (2)          | 393.47 (3)          |
| 1974HeYW <sup>+</sup>          | 91.31 (3)          | 93.32 (2)          | 184.56 (2)          | 208.93 (2)          | 300.186 (20)        | 393.484 (30)        |
| 1977Ab02                       |                    |                    | 184 (1)             |                     |                     | 393 (3)             |
| 1978Du04                       |                    | 93.30 (5)          | 184.63 (3)          | 208.91 (4)          | 300.24 (5)          | 393.54 (5)          |
| 1978Me10 <sup>‡</sup>          | 91.266 (5)         | 93.311 (5)         | 184.577 (10)        | 208.951 (10)        | 300.219 (10)        | 393.529 (10)        |
| 1990Me15                       | 91.237 (35)        | 93.291 (30)        | 184.569 (30)        | 208.97 (3)          | 300.230 (25)        | 393.539 (25)        |
| 2015Ch57                       | 91.41 (5)*         | 93.44 (5)*         | 184.59 (5)          | 208.95 (5)          | 300.20 (5)          | 393.51 (6)          |
| <b>Evaluated<sup>§</sup></b>   | <b>91.270 (14)</b> | <b>93.307 (12)</b> | <b>184.577 (13)</b> | <b>208.940 (14)</b> | <b>300.208 (14)</b> | <b>393.530 (16)</b> |
| $\chi^2/\chi^2_{\text{crit.}}$ | 1.08 / 3.32        | 0.54 / 3.02        | 1.48 / 2.41         | 0.39 / 3.02         | 0.54 / 3.02         | 1.03 / 2.64         |

\* Excluded by the Chauvenet criterion

<sup>‡</sup> Superseded by [1990Me15]

<sup>+</sup> Weighted mean of the two values with minimum experimental uncertainty

<sup>§</sup> Weighted mean and internal uncertainty, except for  $\gamma_{2,1}$  (external uncertainty)

Table 6 – Evaluated  $\gamma$ -ray and recoil energies emitted in  $^{67}\text{Cu}$  decay.

| Emission       | Energy (keV) | Recoil (eV)   |
|----------------|--------------|---------------|
| $\gamma_{2,1}$ | 91.270 (14)  | 0.066827 (21) |
| $\gamma_{1,0}$ | 93.307 (12)  | 0.069843 (18) |
| $\gamma_{2,0}$ | 184.577 (13) | 0.273306 (38) |
| $\gamma_{3,2}$ | 208.940 (14) | 0.350217 (47) |
| $\gamma_{3,1}$ | 300.208 (14) | 0.72300 (7)   |
| $\gamma_{3,0}$ | 393.530 (16) | 1.24237 (10)  |

Finally, the adopted  $\gamma$ -ray energies and level energies of  $^{67}\text{Zn}$  have been determined from a least-square fitting on the evaluated energies, realized with the GTOL program. Table 7 summarizes the adopted values.

Table 7 – Adopted  $\gamma$ -ray and level energies from least-square fitting.

| Emission       | Energy (keV) | Level | Energy (keV) |
|----------------|--------------|-------|--------------|
| $\gamma_{2,1}$ | 91.270 (10)  | 0     | 0            |
| $\gamma_{1,0}$ | 93.310 (9)   | 1     | 93.310 (9)   |
| $\gamma_{2,0}$ | 184.579 (10) | 2     | 184.579 (10) |
| $\gamma_{3,2}$ | 208.943 (10) |       |              |
| $\gamma_{3,1}$ | 300.212 (10) |       |              |
| $\gamma_{3,0}$ | 393.521 (10) | 3     | 393.522 (10) |

### 1.3.2. Relative $\gamma$ -ray emission intensities

Some relative  $\gamma$ -ray emission intensities have been identified in the literature but have not been considered in the present evaluation. The value from [1954Bo97] for the 93-keV emission has been ignored in absence of any reported uncertainty. The values from [1957Be59] have been excluded by the Chauvenet criterion; in addition, the 91- and 93-keV peaks were not resolved in this experiment. The values from [1967Vr03] given in [1969Ra15] have not been considered in absence of the 184-keV reference peak.

Table 8 presents the consolidated datasets and the adopted values. All the datasets are consistent ( $\chi^2$  and  $\chi^2_{\text{crit.}}$  are provided in Table 8). Each adopted relative intensity corresponds to the weighted average with internal uncertainty.

Table 8 – Measured relative  $\gamma$ -ray emission intensities.

| Reference                      | $\gamma_{2,1}$    | $\gamma_{1,0}$    | $\gamma_{2,0}$ | $\gamma_{3,2}$     | $\gamma_{3,1}$    | $\gamma_{3,0}$   |
|--------------------------------|-------------------|-------------------|----------------|--------------------|-------------------|------------------|
| 1969Ra15                       | 15.5 (11)         | 36.0 (25)         | 100 (7)        | 0.24 (3)           | 1.57 (12)         | 0.43 (4)         |
| 1974HeYW                       | 15.5 (10)         | 34.5 (20)         | 100 (2)        | 0.24 (2)           | 1.64 (15)         | 0.48 (6)         |
| 1978Me10                       | 14.37 (21)        | 33.06 (41)        | 100.00 (62)    | 0.236 (10)         | 1.637 (23)        | 0.452 (16)       |
| 2015Ch57                       | 14.23 (14)        | 33.5 (3)          | 100            | 0.243 (5)          | 1.68 (3)          | 0.448 (11)       |
| <b>Adopted</b>                 | <b>14.30 (11)</b> | <b>33.39 (24)</b> | <b>100</b>     | <b>0.2416 (43)</b> | <b>1.650 (18)</b> | <b>0.449 (9)</b> |
| $\chi^2/\chi^2_{\text{crit.}}$ | 1.00 / 3.79       | 0.73 / 3.79       |                | 0.12 / 3.79        | 0.60 / 3.79       | 0.18 / 3.79      |

### 1.3.3. Mixing ratios and multipolarities

Table 9 – Mixing ratios and multipolarities adopted from  $^{67}\text{Ga}$  evaluation [2013BeZP].

| Emission       | Energy (keV) | $ \delta $ | Multipolarity     |
|----------------|--------------|------------|-------------------|
| $\gamma_{2,1}$ | 91.270 (10)  | 0.123 (25) | M1 + 1.5(6)% E2   |
| $\gamma_{1,0}$ | 93.310 (9)   |            | E2                |
| $\gamma_{2,0}$ | 184.579 (10) | 0.31 (7)   | M1 + 8.8(36)% E2  |
| $\gamma_{3,2}$ | 208.943 (10) | 0.042 (17) | M1 + 0.18(14)% E2 |
| $\gamma_{3,1}$ | 300.212 (10) | 0.178 (10) | M1 + 3.07(33)% E2 |
| $\gamma_{3,0}$ | 393.521 (10) | 0.051 (16) | M1 + 0.26(16)% E2 |

The mixing ratios and multipolarities have been adopted from the  $^{67}\text{Ga}$  DDEP evaluation [2013BeZP]. No subsequent measurement has been documented in the extant literature. Table 9 summarizes the adopted values.

#### 1.3.4. Internal conversion coefficients

Table 10 summarizes the measured internal conversion coefficients available in the literature. The datasets are not sufficiently complete for ensuring a robust evaluation. The internal conversion coefficients have thus been calculated with, and adopted from the BrIcc code [2008Ki07]. The values, given in Table 11, are in reasonable agreement with the measurements from [1966Fr12, 1969Li04].

Table 10 – Measured internal conversion coefficients.

| Reference | $\gamma_{2,1}$                    | $\gamma_{1,0}$               | $\gamma_{2,0}$                   | $\gamma_{3,2}$                    | $\gamma_{3,1}$                   | $\gamma_{3,0}$                |
|-----------|-----------------------------------|------------------------------|----------------------------------|-----------------------------------|----------------------------------|-------------------------------|
| 1938Al02  |                                   | $\alpha_T = 0.7$             |                                  |                                   |                                  |                               |
| 1953Ea11  |                                   | $\alpha_T = 0.5$ (2)         | $\alpha_T = 0.012$ (2)           |                                   |                                  |                               |
| 1953Ke51  | $\alpha_K = 0.074$                | $\alpha_K = 0.63$            | $\alpha_K = 0.011$               | $\alpha_K = 0.029$                | $\alpha_K = 2.9 \text{ e-}3$     | $\alpha_K = 1.9 \text{ e-}3$  |
| 1953Me52  |                                   | $\alpha_T = 0.54$ (5)        |                                  |                                   |                                  |                               |
| 1966Fr12  |                                   | $\alpha_K =$<br>0.77 (8)     | $\alpha_K =$<br>0.0156 (10)      | $\alpha_K =$<br>0.0075 (7)        | $\alpha_K =$<br>3.37 (3) e-3     | $\alpha_K =$<br>1.92 (15) e-3 |
|           |                                   | $\alpha_{L1} =$<br>0.073 (8) | $\alpha_{L1} =$<br>1.58 (12) e-3 | $\alpha_{Ltot} =$<br>7.1 (10) e-4 | $\alpha_{Ltot} =$<br>3.5 (3) e-4 |                               |
|           |                                   | $\alpha_M =$<br>0.0132 (18)  | $\alpha_M =$<br>2.4 (3) e-4      | $\alpha_M =$<br>1.23 (3) e-4      | $\alpha_M =$<br>4.5 (7) e-5      |                               |
| 1969Li04  | $\alpha_K =$<br>0.066 (10)        |                              |                                  |                                   |                                  |                               |
|           | $\alpha_{L1} =$<br>6.9 (11) e-3   |                              |                                  |                                   |                                  |                               |
|           | $\alpha_{L2,3} =$<br>6.9 (30) e-4 |                              |                                  |                                   |                                  |                               |
| 1988Be55  |                                   |                              | $\alpha_K = 0.89$ (4)            |                                   |                                  | $\alpha_K = 0.883$ (28)       |

Table 11 – Internal conversion coefficient determined with the BrIcc program [2008Ki07].

| ICC             | $\gamma_{2,1}$ | $\gamma_{1,0}$ | $\gamma_{2,0}$ | $\gamma_{3,2}$ | $\gamma_{3,1}$ | $\gamma_{3,0}$ |
|-----------------|----------------|----------------|----------------|----------------|----------------|----------------|
| $\alpha_T$      | 0.091 (6)      | 0.854 (12)     | 0.0169 (21)    | 9.01 (14) e-3  | 3.88 (6) e-3   | 1.93 (3) e-3   |
| $\alpha_K$      | 0.081 (5)      | 0.748 (11)     | 0.0151 (19)    | 8.06 (13) e-3  | 3.48 (6) e-3   | 1.728 (25) e-3 |
| $\alpha_{L1}$   | 7.9 (5) e-3    | 6.67 (10) e-2  | 1.46 (17) e-3  | 7.97 (12) e-4  | 3.42 (6) e-4   | 1.704 (25) e-4 |
| $\alpha_{L2,3}$ | 8.0 (14) e-4   | 2.545 (27) e-2 | 1.19 (24) e-4  | 3.01 (7) e-5   | 1.226 (25) e-5 | 4.40 (6) e-6   |
| $\alpha_{Ltot}$ | 8.7 (7) e-3    | 9.21 (13) e-2  | 1.58 (20) e-3  | 8.27 (13) e-4  | 3.54 (6) e-4   | 1.748 (25) e-4 |
| $\alpha_M$      | 1.25(9) e-3    | 1.300 (19) e-2 | 2.3 (3) e-4    | 1.186 (19) e-4 | 5.08 (8) e-5   | 2.51 (4) e-5   |

### 1.3.5. Absolute $\gamma$ -ray intensities and $\gamma$ transition probabilities

The absolute emission intensity of the 184-keV  $\gamma$ -ray was measured in [2015Ch57] and reported as  $I_{\gamma 2,0} = 44.2$  (6) %. It has served to normalize the relative intensities evaluated in section 1.3.2. Table 12 summarizes the relative and absolute  $\gamma$ -ray emission intensities, the conversion coefficients and the deduced  $\gamma$ -ray transition probabilities, using  $P_{\gamma} = (1 + \alpha_{\tau}) \times I_{\text{abs.}}$ . The  $\beta^{-}$  transition probabilities have then been derived from the balance of the  $\gamma$  transitions (see section 1.2).

Table 12 – Evaluated absolute  $\gamma$ -ray intensities and  $\gamma$  transition probabilities.

| Emission       | Energy (keV) | $I_{\text{rel.}}$ (%) | $I_{\text{abs.}}$ (%) | $\alpha_{\tau}$ | $P_{\gamma}$ (%) |
|----------------|--------------|-----------------------|-----------------------|-----------------|------------------|
| $\gamma_{2,1}$ | 91.270 (10)  | 14.30 (11)            | 6.32 (10)             | 0.091 (6)       | 6.90 (11)        |
| $\gamma_{1,0}$ | 93.310 (9)   | 33.39 (24)            | 14.76 (23)            | 0.854 (12)      | 27.36 (46)       |
| $\gamma_{2,0}$ | 184.579 (10) | 100                   | 44.2 (6)              | 0.0169 (21)     | 44.9 (6)         |
| $\gamma_{3,2}$ | 208.943 (10) | 0.2416 (43)           | 0.1068 (24)           | 0.00901 (14)    | 0.1077 (24)      |
| $\gamma_{3,1}$ | 300.212 (10) | 1.650 (18)            | 0.729 (13)            | 0.00388 (6)     | 0.732 (13)       |
| $\gamma_{3,0}$ | 393.521 (10) | 0.449 (9)             | 0.1985 (48)           | 0.001930 (30)   | 0.1988 (48)      |

## 2 Atomic data

The fluorescence yield data, the relative K X-ray emission probabilities and the ratios  $P(\text{KLX})/P(\text{KLL})$  and  $P(\text{KXY})/P(\text{KLL})$  have been taken from Schönfeld *et al.* [1996Sc06].

The Auger electron and X-ray absolute probabilities have been determined with the EMISSION program [2000Sc47] from the related decay data.

## 3 Consistency

The adopted decay energy  $Q_{\beta} = 560.8$  (8) keV is from [2021Wa16]. The total decay energy released in the  $^{67}\text{Cu}$  decay is calculated using Saisinuc [2008DuZX] as 560.7 (56) keV, in excellent agreement with the adopted  $Q_{\beta}$ .

## 4 References

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|                 |                                                                                                                                                                                                                                    |                                         |
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| <b>1964Al28</b> | D.G. Alkhazov, V.D. Vasilev, G.M. Gusinskii, I.K. Lemberg, V.A. Nabichvrishvili, Izv. Akad. Nauk SSSR, Ser. Fiz. 28, 1683 (1964); Bull. Acad. Sci. USSR, Phys. Ser. 28, 1575 (1965)                                                | [Half-life]                             |
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| <b>1967Vr03</b> | J. Vrzal, B.S. Dzhelepov, J. Liptak, L.N. Moskvina, V.P. Prikhodtseva, B.M. Sabirov, V.G. Tishin, J. Urbanets, L.G. Tsaritsyna, Izv. Akad. Nauk SSSR, Ser. Fiz. 31, 1647 (1967); Bull. Acad. Sci. USSR, Phys. Ser. 31, 1687 (1968) | [I <sub>γ</sub> ]                       |
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| <b>1972La14</b> | F. Lagoutine, J. Legrand, C. Perrot, J.P. Brethon, J. Morel, Int. J. Appl. Radiat. Isotop. 23, 219 (1972)                                                                                                                          | [Half-life]                             |
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| <b>1973Le18</b> | V.E. Lewis, D. Smith, A. Williams, Metrologia 9, 14 (1973)                                                                                                                                                                         | [Half-life]                             |
| <b>1974Ar22</b> | P.O. Aronsson, B.E. Johansson, J. Rydberg, G. Skarnemark, J. Alstad, B. Bergersen, E. Kvale, M. Skarestad, J. Inorg. Nucl. Chem. 36, 2397 (1974)                                                                                   | [E <sub>γ</sub> ]                       |
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